Related work

1. PID control and its extensions

The PID controller is the most used controller in industry [6], [8], [10], and the articles about canals say the same [5]. The first rules for tuning it come from Ziegler and Nichols [9], and in practice the weights are often found by trial. We also found our weights by trial, with a search in the program (chapter 8 in modeling).

- **Anti-windup [7], [11].** When the valve is at its limit, the integral can grow without effect, and this is called windup. The usual solution is to stop adding to the integral at the limit. Our PID+ does this. Our controller limits the integral to ±1500 and it also clears it when the error changes sign. This is different from the usual anti-windup, and in our tests it helps at the start but it hurts when the null position u_0 is not 50.
- **Filtering and noise [13].** Filtered PID and PIDD2 controllers for integrating processes with measurement noise are described in [13]. A tank level is an integrating process. Our smoothing (the average with the previous value) has the same purpose, and our tests show that it is the most important part of our controller.
- **Second derivative (PIDD2) [12], [13], [14].** These articles give tuning rules for controllers with a second derivative, for oscillatory systems with delay [12], for integrating processes [13] and for processes with inertia and delay [14]. The rules start from a model of the process. We use the second derivative too ($\frac{3}{5}$ of the rate and $\frac{2}{5}$ of the acceleration in one derivative part), but we found the weights by search, without a model.

2. Water-level control in canals and reservoirs

The closest problem in the literature is the control of the water level in canals and reservoirs. A classification of the control algorithms for canals is in [16], an overview of the control of irrigation networks is in [17], and the models are in [15]. We took from [15] and from [1] the idea of a tank as an integrator, which is the model of our simulator (chapter 2 of the modeling). The table shows the five articles that we studied in detail.

Article	What it does	Needs a model?	Metrics	Similar to ours	Different from ours
Weyer [1]	PI controllers, one for each pool of a real canal, with a filter and a feedforward term; field tests	Yes, models found by identification are used to design the controllers	Level responses in field tests	PI/PID structure, dead-band on the gate movement (0.015 m), and the outflows to the farmers are not measured,	The design needs a model of each pool

Article	What it does	Needs a model?	Metrics	Similar to ours	Different from ours
				they are disturbances	
Nogueira and Pinho [2]	A nonlinear control law for reservoirs in cascade, with a proof that the level goes to the reference	Yes, the discrete water balance of the reservoirs	Simulations with 2 reservoirs	Steps in discrete time, inflow from rain only estimated and with noise, a minimum change of the flow (like a dead-band)	The control law is not a PID, and it uses the model
Wu et al. [3]	A target interval for the level, a feedforward schedule and a PI-based feedback, tested on a long canal (60 pools and 61 gates)	Yes, the volume balance and the delivery schedule	Number of gate operations, gate openings, levels	It wants fewer gate movements, and it keeps the level in an interval	It uses a schedule of the demand (feedforward), and it is for a real canal
Shahverdi et al. [4]	A PID controller in a hydraulic simulator (HEC-RAS), inside it and from MATLAB, tested on a canal	Yes, the hydraulic model of the simulator	MAE, IAE, efficiency, equity, adequacy	A PID with the weights found by trial and error, IAE as a metric	The process is a hydraulic model of a canal, the controller is a plain PID
Li et al. [5]	A hybrid controller: a feedforward part based on a model and an LQR feedback, on the ASCE Test Canal 2	Yes, a linear model of the canal	MAE, IAE, NIAQ, NIAW, transition time	Hybrid structure (feedforward and feedback), similar error indicators	It needs a model, and it is not a PID. The dead-band is zero in its tests

Table 2.1. The five articles about water-level control

The main difference is the same in all five articles: they use a model of the canal or of the reservoir, or a known or estimated flow. Our controller uses only the level and how it changes between two steps. This

is a possible advantage when the flows are unknown, but it also means that we cannot expect the precision of the controllers that use a model.

3. Salterns

We found only one source about salterns, [18]. It says that a saltern needs a close control of the depths and of the brine transfers, but it does not describe automatic level controllers. We did not find articles with controllers for the water level of a saltern, so we use the canals and reservoirs as the closest case.

4. Metrics in the literature and in our work

Metric	Used in the literature	In our work
IAE or MAE (total error)	[4], [5]	IAE
Maximum error	[5]	Maximum error after the event
Transition or settling time	[5]	Settling time
Overshoot	Classic control [6], [8]	Overshoot (S1 and S3)
Number of gate operations	[3]	Valve movement and direction changes
Efficiency, equity, adequacy, NIAQ, NIAW	[4], [5], for the delivery of water in canals	Not used, they need a canal with users

Table 2.2. The metrics that we use and where they come from

5. Where we expect similar or better results

The table says where we expected similar or better results than the approaches from the literature, and what we found in the simulation (chapter 4).

What we look at	What we expect	What we found
Total error (IAE) at the start and at a change of the set point (S1, S3)	Similar or better than the PID-type controllers [4], because of the nonlinear functions and the second derivative	Better than P, PI, PID and PID+ in both situations
Rain and evaporation (S4)	Similar or better than the PID-type controllers	The smallest IAE of the five controllers
Direction changes of the valve	Fewer than a PID-type controller, because of the smoothing and the dead-band	28 against 198 for PID+ in S1, 48 against 257 in S3, 82 against 234 in S4

What we look at	What we expect	What we found
Total valve movement	Similar to the PID-type controllers. We do not expect to reach the reduction of [3], which uses a schedule	Bigger than PID+ in all the situations, smaller than the plain PID
Unknown change of the flows (S2, S5, S6)	Worse than the controllers with a model [2], [5], and maybe worse than PI and PID+	PID+ is equal or better, and our controller settles less often
Noise on the level reading (S5)	Worse, because the dead-band and the smoothing are small for this noise	Worse than PID+ (22 percent of the runs settled, against 96 percent)
Need of a model or of the flows	A real advantage, because we do not need them	True by design, but it costs precision when the flows change
Absolute values of the errors	We cannot compare, because the processes and the units are different	Not compared

Table 2.3. The expected and the found results

We did not run the controllers from [2] and [5], because they need a model of the process and because they are made for other processes. So the comparison with them is only about the method.

6. Validation on a benchmark

We did not find public data (levels and valve positions) from a saltern. For the next step we propose a benchmark that other articles use. The ASCE Test Canal 2 is described in the test cases for canal control algorithms [19], and it is used in [5], which gives results on it with the indicators MAE, IAE and transition time. The steps would be: to build the canal with several pools in our simulator, to repeat the flow changes of [5], to run P, PI, PID, PID+ and our controller, and to compare the indicators with the ones in [5]. We have not done this yet, so we do not claim any result on a benchmark. A second possibility is to ask a saltern for level and valve data, to run our controller on it in simulation.